



Efficiency analysis of thermoelectric generators

J. Ider*, F.Y. Assahi, A.F. Oliveira, R.M. Rubinger, C.P.L. Rubinger

Universidade Federal de Itajubá, Av BPS, 1303, Pinheirinho, Itajubá, MG 37500-903, Brazil

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ABSTRACT

The negative environmental and economic impact, derived from the use of fossil fuels – oil, coal, gas, and other non-renewable sources of energy – has stimulated scientific research on clean and economically viable sources. Photovoltaics and wind power are intermittent sources which stimulates the quest for other energy sources. In this sense, a promising source of energy is made up of Thermoelectric Generators (TEGs), which have the property of converting thermal energy into electrical energy, through the well-known Seebeck Effect. A new methodology for the characterization of these devices is presented in this work. For a robust analysis of the measures, the extraction of parameters is done through the meta-heuristic Self-Adaptive differential evolution method. This is made possible by simultaneous fitting of three equations and five parameters in a consistent autonomous fashion. Besides a robust experimental setup which allows the generation and precise control of the temperature variable as electrical measurements, in a controlled environment, and a thorough statistical analysis is carried for eight TEGs, model SP-184827145 which gives the mean performance of the manufactured devices. The results obtained with low uncertainties indicates that this methodology is a reliable and low-cost option for the characterization of thermoelectric devices.

1. Introduction

Energy sources are a major concern and growing target for research throughout the scientific community and concern by government agencies, due to their great importance for the social and economic development of the population [1]. Since the economic development of a population is linked to the levels of energy consumption [2], this creates a need to use other options to replace the use of fossil fuels with alternative sources of renewable and sustainable energy [3–6]. Besides, the main motivations are the reduction in reservoirs, fossil fuel price rises, and its impact on the environment.

In recent years, alternative and efficient renewable energy sources have been explored, such as, solar and wind power, which are already part of the energy generation system of several countries [7]. Among several technologies used from these sources, the generation of thermoelectric energy emerges as a great option, being green and clean, which occurs through the use of TEGs [8].

Thermoelectric energy can be generated with the use of TEGs, which are solid state devices. They operate by performing a direct conversion of thermal energy into electrical energy [9]. They have many advantages, such as simple design, non-use of moving parts, greater longevity, without maintenance and operation costs, do not emit harmful pollutants during operation and do not contain chemicals, therefore, do not harm the environment [10–14].

The fabrication of TEGs can be done using different types of technology, a variety of manufacturing methods, and a plurality of materials in the function of the substrate such as silicon [15–18], aluminum [19–23] and polymers [24–26].

Its applications are in diverse and very important areas. Some that we can mention are car engines [27–30], industrial electronic devices [31,32], micro powered wireless platforms [33,34], health monitoring, tracking systems [35–37] and aerospace [38,39].

TEGs are well-studied devices and present in the literature. A general and detailed overview of them was carried out by Nesrini [17], Sidique [40] and Patil [41]. Although these commercial modules are well studied, there is speculation that they may sometimes perform below the specified ratings [42]. This is also due to a difficult experimental analysis of such devices. The main issue is a the necessity of a measurement of three quantities namely electrical thermal conductivities and Seebeck's coefficient. Until recently there was no single setup for measure those quantities and calculate the Figure of Merit [43]. Also, our fitting procedure can lead to consistent values of five parameters considered in three distinct equations in a autonomous fashion, i.e. without any human interference during the process. The seeding of initial parameters and evolution rules are the sole human contribution to this process.

* Corresponding author.

E-mail addresses: joaoider@gmail.com (J. Ider), flavioassahi@gmail.com (F.Y. Assahi), adhimarflavio@unifei.edu.br (A.F. Oliveira), rero@unifei.edu.br (R.M. Rubinger), carlalar@unifei.edu.br (C.P.L. Rubinger).

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Nomenclature

TEG	Thermoelectric Generator
TEC	Thermoelectric Cooler
DE	Differential Evolution
SADE	Self-Adaptive Differential Evolution
α	Seebeck coefficient
I	Current
I_{max}	Maximum current
I_{sc}	Short circuit current
R	Internal Resistance
R_L	Load Resistance
V	Voltage
V_{oc}	Open circuit voltage
T_h	Hot temperature
T_c	Cold temperature
$\bar{T}$	Average temperature
κ	Thermal conductivity
ρ	Electrical resistivity
η	Thermal efficiency
η_c	Thermal efficiency of Carnot machine
η_{max}	Maximum efficiency
W	Power dissipation
W_{max}	Maximum power
Z	Z-factor
Z_{eff}	Effective Z-factor
ZT	Figure of Merit
N	Number of <i>pn</i> junctions
$P_{i,j}$	Initial population
G	Number of iterations
D_i	Donator vector
F	Mutation Factor
$T_{i,j}$	Crossover Vector
CR	Crossing rate

Because of these difficulties in presenting robustly the parameters that demonstrate the efficiency of thermoelectric devices, we present in this work a new methodology that proves to be more efficient. For this purpose, eight thermoelectric energy generators (TEGs) were used to perform measurements, through a new and efficient experimental arrangement, which generates and precisely controls a temperature gradient while performing electrical measurements. In addition, further analysis of these data is presented through a metaheuristic method, Self-adaptive Differential Evolution (SADE), algorithm, first introduced by Storn and Price [44], is a population-based stochastic function optimization algorithm [45], which optimizes a function by following evolutionary rules. It is a powerful tool to find the global minimum of non-linear and multi-modal functions and was used previously in power generation devices as solar cells [46]. It is thoroughly described in the work of Febba and cols. The use of such methods allow simultaneously to fit current, voltage and power curves with precise parameters estimation that help define the performance of solid state power generate devices as TEGs subject of this research. For solar cells that is a vast literature on distinct methods for parameters estimation but for TEGs the applied methods are not main issue of the research. Therefore our choice of methods was based on one of the most promising methods for estimate solar cells parameters. For our knowledge, previous works have not considered simultaneous fittings which is one of our main contributions.

In summary this work tested the contributions of both the fitting method of the SADE algorithm together with a robust setup form

the analysis of commercial thermoelectric generators. We also provide efficiency data in a similar fashion to photovoltaic devices which allows the comparison with other technologies. Our results could establish a new approach for characterizing commercial devices for the purpose of defining the optimal parameters and ratings for the datasheet of thermoelectric power generating commercial devices.

2. Review of thermoelectric devices

The modeling of the experimental data is done through the theoretical models of the TEGs [47–50] and, in our analysis, we first looked at the power delivered by the device. This power is a function of the number of *pn* junctions *N*, Seebeck coefficient α , current *I*, hot temperature T_h , cold temperature T_c , internal resistance *R* and load resistance R_L . The power of the TEG does not depend directly on the current or the voltage and can be expressed as

$$W = \frac{N\alpha^2(T_h - T_c)^2}{R} \frac{R_L/R}{(1 + R_L/R)^2}. \quad (1)$$

The voltage as a function of *N* is

$$V = \frac{N\alpha(T_h - T_c)}{\frac{R_L}{R} + 1} \frac{R_L}{R}, \quad (2)$$

and the current is

$$I = \frac{\alpha(T_h - T_c)}{R_L + R}. \quad (3)$$

Another important parameter for thermoelectric generators is their efficiency. One can calculate the thermal efficiency of an ideal Carnot machine, given by

$$\eta_c = 1 - \frac{T_c}{T_h}, \quad (4)$$

which can be compared with the thermal efficiency obtained from the ratio between the power and the heat absorbed by the hot part, which is presented in the [Appendix](#), Eq. (16).

Finally, we show here the expressions relative to the theoretically calculated effective quantities, important for TEG analysis. We have, first, the effective resistivity

$$\rho = \frac{4W_{max}A/L}{N(I_{max})^2}, \quad (5)$$

and the effective Z-factor

$$Z_{eff} = \frac{1}{\bar{T}} \left[\left(\frac{1 + \frac{T_c \eta_{max}}{T_h \eta_c}}{1 - \frac{\eta_{max}}{\eta_c}} \right)^2 - 1 \right]. \quad (6)$$

All these quantities can be calculated at their maximum point. Expressions for W_{max} , V_{max} , I_{max} , η_{max} are given in [Appendix](#).

Thus, this work consists of the experimental measurements of a Thermoelectric Generator and its subsequent analysis robustly by the metaheuristic method named Self-adaptive Differential Evolution, considering the equations presented here that were used in a simultaneous fitting procedure and will be described in the following section.

3. Materials and methods

3.1. Differential evolution algorithm

Genetic algorithms are a particular type of evolutionary algorithms that use approaches derived from evolutionary biology such as heredity, mutation, natural selection and recombination [51]. They are like a computer simulation where a fixed amount of possible solutions, called a population, is evolved in search of better, more adapted solutions. This evolution starts from a population created randomly and goes through several mutations, or processes, through its generations. With each new generation, the new set of solutions is evaluated, and its

Table 1
Sample dimensional parameters, number of pn junctions and Z factor.

L (cm)	A (cm ²)	N	Z
0.2	0.01	128	0.00209

adaptation to the environment is verified (in this case, the adaptation to some mathematical criteria previously established, called the objective function). The most adapted are then selected for the next generation, recombined, mutated, and form a new population. This new population is then used as an entry for the next iteration until a pre-defined criterion is established.

In this work, the experimental data collected were then analyzed according to a genetic algorithm, namely, the method of differential self-adaptive evolution [46,52]. This algorithm, given the data provided in this study, provides us with the Seebeck coefficient values (α), internal resistance (R), and the hot and cold temperatures (T_h e T_c). In order to obtain high quality consistent parameters from a fitting procedure, Eqs. (1)–(3) must be fitted simultaneously in a feedback process that allows precise estimation of the short circuit current (I_{sc}), open-circuit voltage (V_{oc}), maximum power (W_{max}), resistivity (ρ) of Eq. (5), efficiency (η) of Eq. (4) and the effective Z-factor (Z_{eff}) of Eq. (6) of the TEG modules studied.

The metaheuristic method, called Differential Evolution (DE), was proposed by Storn and Price [44]. It had its development done quickly, already being effective for a large number of problems [53–56], considering that at the beginning it was applied to solve systems of continuous variables, only [44]. One of algorithm characteristics is its independence on the control parameters, which turns it in a self-adaptive procedure [57], and for that reason, its coined as Self-adaptive Differential Evolution.

For the carrying out the calculation of the parameters of the thermoelectric generators it is important to consider their constitutive and intrinsic properties. These properties are L, the characteristic length of a single junction pn , A, the cross-sectional area, N, the number of junctions pn and Z, the Z-factor. These values are shown in the Table 1.

First, for the generation of solutions, a set of individuals is randomly generated. These individuals, potential solutions, will evolve over the generations. Each of these individuals is a candidate to be the best adapted to the environment, that is, to minimize the objective function, a mathematical criterion previously established. This initial population is obtained through Eq. (7)

$$P_{i,j}^{G=1} = P_i^{G=1} + rand_{i,j}(0,1)(P_u^{G=1} - P_l^{G=1}), \quad (7)$$

where $rand_{i,j}(0,1)$ is a random real number, which belongs to the range between 0 and 1. The index i represents the size of the population NP (in this case 100), and the index j , ranging from 1 to 4, the number of parameters studied, which are – as stated earlier – Seebeck coefficient (α), internal resistance (R) and hot and cold temperatures (T_h e T_c). P_u and P_l are the upper and lower limits that define the so-called search space. G represents each new generation of populations, ranging from 1 to $G_{max}=50000$, the value chosen for this study. Each P_i is a candidate to be an optimal solution, the most adapted.

After the generation of the initial population, the mutation stage occurs, where a donor vector is generated, and obtained through Eq. (8)

$$D_i^G = P_i^G + F(P_{best}^G - P_i^G) + F(P_{r1}^G - P_{r2}^G), \quad (8)$$

where D is the mutation vector that will select the tested individuals. (G, P_i^G) is the generation of individuals where P_i^G represents the current individual. (P_{r1}^G and P_{r2}^G) are chosen at random, exactly to be different from P_i^G . P_{best}^G is the best individual, the most adapted, the one who minimizes the objective function f.

One of the self-adapting values in this algorithm is F, a real number, called a mutation factor and which belongs to the range ($F \in [0, 2]$). For each iteration, its value is adopted by Eq. (9)

$$F_i^{G+1} = \begin{cases} F_l + rand_1 F_u, & \text{if } rand_2 < \tau_1, \\ F_i^G, & \text{otherwise.} \end{cases} \quad (9)$$

The next stage of evolution is called a crossover. It is defined by mixing the target vector P_i^G , produced by a vector T_i^G between 0 and 1, and carries a comparison between the crossover values as:

$$T_{i,j}^G = \begin{cases} D_{i,j}^G, & \text{if } rand(0,1)_j \leq C \text{ or } j = j_{rand}, \\ P_{i,j}^G, & \text{otherwise.} \end{cases} \quad (10)$$

Here, $D_{i,j}^G$ is the donor vector, j_{rand} is a random integer between 1 and 6 (size of the search space). C is another self-adaptive factor in the algorithm. It is also a real number, now belonging to the range ($C \in [0, 1]$) and called the crossover rate. It is defined by Eq. (11)

$$C_i^{G+1} = \begin{cases} rand_3, & \text{if } rand_4 < \tau_2, \\ C_i^G, & \text{otherwise.} \end{cases} \quad (11)$$

In this equation, $rand_4$ is 4 random numbers between 0 and 1. τ_1 and τ_2 represent the probability of adjusting the mutation factor F and the crossover rate C.

Finally, in the selection step, the temporary vector and the P_i^G vector are compared and the vector that minimizes the objective function is chosen, using Eq. (12)

$$P_i^{G+1} = \begin{cases} T_i^G, & \text{if } f(T_i) < f(P_i), \\ P_i^G, & \text{otherwise.} \end{cases} \quad (12)$$

This process is repeated until the end of the generations, or when some pre-defined criteria are satisfied. In the case of this work, the evolution of the solutions is finalized when reaching the 50000 generation. Thus, a simultaneous fitting of the variables V, I, and W is completed.

The parameters associated with the experimental curves are shown in the Tables 3 and 6, in the results section.

3.2. Experimental results

The measurements were taken at the Laboratório de Caracterização de Materiais (LCM) of the Instituto de Física e Química (IFQ) of Universidade Federal de Itajubá (UNIFEI). Eight Bismuth Telluride (Bi_2Te_3) TEGs, model SP-184827145 were used individually, to obtained better statistics. They withstands temperatures up to 200 °C, with length 40 mm, width 40 mm and height 3.4 mm, insulated with Teflon and connected with copper cables [42,58].

Each TEG operates with a temperature gradient between its hot and cold sides which generates a voltage drop between those sides through the Seebeck effect. Since the generated voltage drop is proportional to the temperature gradient and the proportionality constant as the module's Seebeck coefficient, measurements needs to be carried out through specific experimental setup shown in Fig. 1 comprising The thermoelectric generator is divided into 3 parts: Cooling System, Control System and Voltmeters. The Cooling System consists of a Cold Water Reservoir and through a digital temperature controller and BK Precision1786B it is possible to cool the material. The control system consists of BK PrecisionXLN 3640, ArduinoMega2560, Arduino Uno and Sensor Max 6675. The voltmeters used are Keithley 2100, BK Precision 8500 and BK Precision 9202. a power source (BK precision 9202) used to oppose the generated module's voltage, and electronic load (BK precision 8500) to dissipate the generated power and a voltmeter (Keithley 2100) to acquire module's voltage and current as I(V) characteristics and this defines the measurement circuit. However in order to produce reliable and stable results we developed a control system to be used together with the measuring circuit with the sole function to assure stable temperature gradient between hot and cold sides of the module.

The Temperature Control Unit comprised a sub-setup and is used to control the TEG module's temperature. It is composed by a Microcomputer temperature controller, model UYUE 946S UY-D220V as the heat constant temperature source T_h (limited the control to up to 150 °C) with the hot TEG module's face placed in direct contact with the hot side of the UYUE 946S UY-D220V. The cold TEG module's face was put in contact with the cold side of a Peltier Thermoelectric

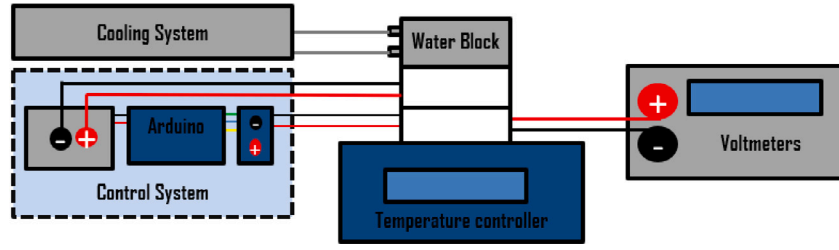


Fig. 1. Connection scheme used in the experimental apparatus for measuring curves $I - V$.

Cooler TEC1-12706 to act as the constant cold temperature T_c . The TEC1-12706 however needs to be connected to a heat sink in order to keep T_c really stable. This is carried out by putting a metallic water block placed on the top of TEC1-12706 hot side and through which cold water circulates and removes the heat generated by the TEC1-12706 Peltier cooler. To finish the cold side control circuit, this water block was connected by hoses to a cold-water reservoir kept at cold with water and ice. The circulation of this water, between the reservoir and the water blocks, was made by a BF12 fine nozzle water pump, which was supplied with 5 V by a BK Precision 1786B source. The water temperature measurement was carefully monitored by a Digital Controller Temperature KT1210 W.

The control of T_c is carried just by controlling the power driven to the TEC-12706 Peltier cooler. The control of the voltage applied to the Peltier TEC was done through a standard Proportional Integral and Derivative (PID) routine developed in Python, which regulated the applied voltage and kept T_c stable and constant throughout one $I(V)$ measurement. This routine controlled a voltage output of an Arduino Mega 2560, through its PWM (Pulse Width Modulation) port which was in turn connected to the voltage outputs of a BK Precision XLN 3640 power supply that drives the TEC-12706 Peltier cooler. The pulsed PWM output needs to be converted to continuous voltage prior to its connection with the voltage control port of the XLN 3640 and this process is performed by a low-pass filter (built in the laboratory), connected to the Arduino Mega 2560 PWM port.

The characterization of the TEG's device and its generated power is carried out by $I(V)$ measurements considering some fixed generated temperature gradients of 10 °C, 20 °C, 30 °C, 40 °C, 50 °C, 70 °C, 80 °C and 90 °C. The cold temperature T_c was kept at 20 °C. The measurements carried with the BK Precision 9202 Multi-Range DC Power Supply, BK Precision 8500 Programmable DC Electronic Load and digital multimeter Keithley 2100 6 1/2 Digit were controlled by a LabView program that changes the applied voltage in 5 mV steps from 0 V to the open circuit voltage observed for each temperature gradient. Temperature T_c and T_h were also measured during experiments using two MAX 6675 Sensors, which use type K thermocouples. These measured the TEG's cold and hot temperatures. The MAX 6675 Sensors were connected to an Arduino Uno for data acquisition in the developed software.

For each of the eight TEGs, nine measurements were made, where gradients of approximately, 10 °C, 20 °C, 30 °C, 40 °C, 50 °C, 70 °C, 80 °C and 90 °C.

For each of the seventy two measurements, SADE was used to analyze and extract parameters. The method was operated with an initial of 100 individuals for each parameter, evolving them for 50000 generations, and running each program 40 times, with a random initial population. For each set of 40 solutions, a statistical analysis was performed which provided us with the parameters of hot temperature T_h , cold temperature T_c , Seebeck coefficient α , resistance R from the studied model of Eqs. 1 to (6), short circuit current could be calculated I_{sc} , open-circuit voltage V_{oc} , maximum power W_{max} , voltage at maximum power $V(W_{max})$, current at maximum power $I(W_{max})$, resistivity ρ , Carnot efficiency η_c , maximum efficiency η_{max} and effective Z-factor Z_{eff} .

Table 2

Medians of temperature gradients ΔT (°C), Seebeck coefficient α (μVK^{-1}) and resistance R (Ω).

	ΔT (°C)	α (μVK^{-1})	R (Ω)
1	11 ± 1	0.30 ± 0.03	0.01799
2	20 ± 1	0.33 ± 0.02	0.01929
3	30 ± 2	0.34 ± 0.02	0.02025
4	40 ± 1	0.35 ± 0.01	0.02145
5	50 ± 2	0.33 ± 0.01	0.02213
6	60 ± 2	0.35 ± 0.01	0.02468
7	71 ± 2	0.34 ± 0.01	0.02571
8	81 ± 2	0.34 ± 0.01	0.02679
9	94 ± 3	0.35 ± 0.01	0.02885

Table 3

Medians of the short circuit current values I_{sc} , open-circuit voltage V_{oc} , maximum power W_{max} , voltage at maximum power $V(W_{max})$ and current at maximum power $I(W_{max})$.

	I_{sc} (A)	V_{oc} (V)	W_{max} (W)	$V(W_{max})$	$I(W_{max})$
1	0.17 ± 0.02	0.40 ± 0.06	0.01772	0.19793	0.08372
2	0.33 ± 0.02	0.84 ± 0.07	0.06988	0.42134	0.16674
3	0.49 ± 0.03	1.32 ± 0.08	0.15622	0.65083	0.24403
4	0.61 ± 0.03	1.71 ± 0.07	0.24890	0.84866	0.30622
5	0.73 ± 0.03	2.06 ± 0.09	0.37332	1.03186	0.36514
6	0.85 ± 0.03	2.61 ± 0.09	0.54331	1.30286	0.42071
7	0.94 ± 0.03	3.05 ± 0.11	0.68436	1.51865	0.46263
8	1.03 ± 0.04	3.49 ± 0.14	0.87320	1.75340	0.51318
9	1.10 ± 0.04	4.09 ± 0.15	1.08757	2.04016	0.55104

4. Results and discussion

Each TEG then had nine measurements, varying the temperature gradient from 10 °C to 90 °C, with steps of 10 °C. With that, all the equivalent measures between the TEGs were used and statistical results of the parameters that determine the functioning of the TEGs are presented here. In Table 2 the results of measure 1 are presented, which refer to a statistical analysis of the measures with gradients of 10 °C of each TEG, measure 2 with gradients of 20 °C, successively up to the measure 9, which is the statistical analysis of all measurements performed with a gradient of 90 °C. Thus, the Table 2 presents the median values of the temperature gradients ΔT (°C), Seebeck coefficient α (μVK^{-1}) and resistance R (Ω). Although the set of TEGs was acquired in the same batch, it is possible to verify a great variation in the values of the parameters, which confirms what was observed by Saim et al. [42].

In Table 3 we present medians of short circuit current values I_{sc} , open-circuit voltage V_{oc} , maximum power W_{max} , voltage at maximum power $V(W_{max})$ and current at maximum power $I(W_{max})$. The values are consistent with the experiment, increasing with increasing temperature gradient between the faces. However, we already see a noticeable difference here.

The values provided by the devices' datasheet are shown in Table 4. This shows, for a given temperature gradient, an associated open-circuit voltage.

A comparison is then made between the theoretical (provided by the datasheet) and the experimental (from our research) values of the open-circuit voltage for a given temperature gradient. A variation in these

Table 4
Values provided by the device's datasheet.

ΔT (°C)	V_{oc} (V)	I (mA)
20	0.97	225
40	1.8	368
60	2.4	469
80	3.6	558
100	4.8	669

Table 5

Comparison between theoretical and experimental values of open-circuit voltage for temperature gradient data.

ΔT (°C)	V_{oc} (V) theoretical	V_{oc} (V) experimental
20	0.97	0.84 ± 0.07
40	1.8	1.71 ± 0.07
60	2.4	2.61 ± 0.09
80	3.6	3.49 ± 0.14

Table 6

Medians of the resistivity values ρ , Carnot efficiency η_c , maximum efficiency η_{max} and effective Z-factor Z_{eff} .

	ρ (m Ω m)	η_c (%)	η_{max} (%)	Z_{eff} (K ⁻¹)
1	0.91 ± 0.24	53.35 ± 3.44	0.22 ± 0.04	0.00209
2	0.97 ± 0.16	94.85 ± 2.73	0.91 ± 0.09	0.00209
3	1.01 ± 0.11	94.75 ± 1.74	1.37 ± 0.09	0.00209
4	1.07 ± 0.09	91.05 ± 1.23	1.69 ± 0.08	0.00209
5	1.11 ± 0.09	92.45 ± 1.25	2.22 ± 0.10	0.00209
6	1.22 ± 0.09	87.75 ± 0.88	2.35 ± 0.09	0.00209
7	1.26 ± 0.10	86.45 ± 0.85	2.58 ± 0.10	0.00209
8	1.35 ± 0.12	80.50 ± 0.70	2.66 ± 0.10	0.00209
9	1.47 ± 0.13	76.25 ± 0.68	2.97 ± 0.11	0.00209

values is noted here, even for TEGs in the same batch. These data can be seen in Table 5. The experimental values is close to the theoretical V_{oc} but lower due to contact resistance or chemical potential mismatch between layers of the device.

The medians of the resistivity values ρ , Carnot efficiency η_c , maximum efficiency η_{max} and effective Z-factor Z_{eff} are shown in Table 6. The results obtained for the effective Z-factor of the devices were exactly those presented in the literature, without alteration for all the studied TEGs. Efficiency, as expected, increased with increasing temperature gradient, reaching 2.97% for a gradient of 90 °C. Table 6 main message is of a fixed Z_{eff} with all the variations the second law of thermodynamics as in the operation of an idealized Carnot machine. The experimental errors or uncertainties are resulting from two statistical process: The first due to the fitting itself and the second from the behavior of eight distinct modules.

The Fig. 2 presents the values of current per voltage and power per voltage showing the experimental results (in red) and those generated from the proposed model (in black). Curves 1, 2, 3, 4, 5, 6, 7, 8 and 9 represent, respectively, the values for the temperature gradients of 10 °C, 20 °C, 30 °C, 40 °C, 50 °C, 60 °C, 70 °C, 80 °C and 90 °C.

We see a linear behavior in the I(V) curves and, as you have larger temperature gradients, the experimental values fluctuate more. This is due to the TEG's behavior and the nature of the electrical measurement performed in this experiment. When generating a temperature gradient in the TEG, it provides a voltage value that passes through the load resistance, here represented by the electronic load BK precision 8500. This resistance, in turn, makes this voltage pass back into the TEG, causing a disturbance in its temperature.

This same behavior is seen in Fig. 2, on the W(V) curve, where there are greater oscillations in the experimental values for greater temperature gradients.

Analyzing the data presented in Fig. 2, we see that the experimental arrangement created for this experiment was satisfactory. He fulfilled his goal of generating temperature gradients and controlling them, in addition to measuring them, a fact observed in the graphs (except

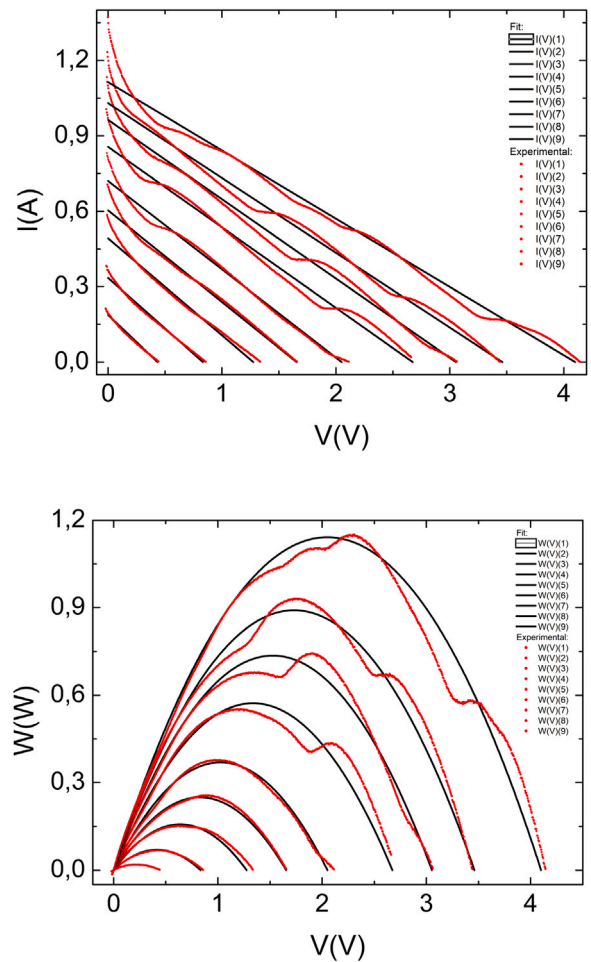


Fig. 2. (a) Current by voltage and (b) voltage power for experimental values (in red) and generated from the proposed model (in black). Curves 1, 2, 3, 4, 5, 6, 7, 8 and 9 represent, respectively, the values for the temperature gradients of 10 °C, 20 °C, 30 °C, 40 °C, 50 °C, 60 °C, 70 °C, 80 °C and 90 °C.

the oscillations previously discussed). SADE's good behavior was also noted, allowing a consistent fitting process and statistics not only from distinct TEGs but in comparison between initial parameters that were randomly generated.

5. Summary and conclusions

In this work we have presented a new methodology for analyzing thermoelectric generating devices that consists in an autonomous and simultaneous fitting of I , V and W that allow the precise determination of relevant associated electrical quantities. That procedure could not be carried without considering our proposed experimental arrangement, which generates and controls temperature gradients with precision, measures temperatures that have assured that the relevant variables were at stable thermodynamic conditions. The results from this assembly lead to acceptable statistical variations, compatible with the expected for commercial devices. The applied method of data analysis through a genetic algorithm of differential evolution was used for the first time for thermoelectric generators and presented excellent results qualifying it as a powerful tool for the analysis of thermoelectric materials. The autonomous fitting process with simultaneous equations and parameters being analyzed lead to a successful method to obtain precise performance ratings of thermometric generators. The TEGs studied, despite showing variations in their characteristics between elements of

the same batch behaved reliably, presenting a good thermal efficiency, from 2.97% to a temperature gradient of 90 °C. This evidence the robustness of the proposed methodology to define precisely relevant operation ratings for commercial thermoelectric generators.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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Appendix

Making $R_L = R$ [59] we get the maximum quantities. Thus, the power described in Eq. (1) is

$$W_{max} = \frac{N\alpha^2(T_h - T_c)^2}{4R}, \quad (13)$$

while the current in Eq. (3) is

$$I_{max} = \frac{\alpha(T_h - T_c)}{2R} \quad (14)$$

and the voltage of Eq. (2)

$$V_{max} = \frac{N\alpha(T_h - T_c)}{2}. \quad (15)$$

The efficiency is

$$\eta = \frac{\left(1 - \frac{T_c}{T_h}\right) \frac{R_L}{R}}{\left(1 + \frac{R_L}{R}\right) - \frac{1}{2} \left(1 - \frac{T_c}{T_h}\right) \frac{\left(1 + \frac{R_L}{R}\right)^2 \frac{T_c}{T_h}}{ZT_c}}, \quad (16)$$

where $ZT_c = \frac{\alpha^2 T_c}{\kappa \rho}$ is the Figure of Merit (ZT), κ the thermal conductivity, and ρ the electrical resistivity. and the maximum efficiency

$$\eta_{max} = \left(1 - \frac{T_c}{T_h}\right) \frac{\sqrt{1 + Z\bar{T}} - 1}{\sqrt{1 + Z\bar{T}} + \frac{T_c}{T_h}}, \quad (17)$$

where $\bar{T}$ is the average temperature between the hot and cold sides, expressed by

$$\bar{T} = \frac{T_h + T_c}{2}. \quad (18)$$

References

- [1] V. Pandiyarajan, M. Chinna Pandian, E. Malan, R. Velraj, R.V. Seeniraj, Experimental investigation on heat recovery from diesel engine exhaust using finned shell and tube heat exchanger and thermal storage system, *Appl. Energy* 88 (1) (2011) 77–87.
- [2] R. Saidur, M. Rezaei, W.K. Muzammil, M.H. Hassan, S. Paria, M. Hasanuzzaman, Technologies to recover exhaust heat from internal combustion engines, *Renew. Sustain. Energy Rev.* 16 (8) (2012) 5649–5659.
- [3] Hongxia Xi, Lingai Luo, Gilles Fraisse, Development and applications of solar-based thermoelectric technologies, *Renew. Sustain. Energy Rev.* 11 (5) (2007) 923–936.
- [4] Abdeen Mustafa Omer, Focus on low carbon technologies: The positive solution, *Renew. Sustain. Energy Rev.* 12 (9) (2008) 2331–2357.
- [5] O. Afshar, R. Saidur, M. Hasanuzzaman, M. Jameel, A review of thermodynamics and heat transfer in solar refrigeration system, *Renew. Sustain. Energy Rev.* 16 (8) (2012) 5639–5648.
- [6] Mirunalini Thirugnanasambandam, S. Iniyar, Ranko Goic, A review of solar thermal technologies, *Renew. Sustain. Energy Rev.* 14 (1) (2010) 312–322.
- [7] Hannah Ritchie, Max Roser, *Renewable energy*, Our World in Data (2020) <https://ourworldindata.org/renewable-energy>.
- [8] D.M. Rowe, Thermoelectrics, an environmentally-friendly source of electrical power, *Renew. Energy* 16 (1–4) (1999) 1251–1256.
- [9] Antonino Proto, Daniele Bibbo, Martin Cerny, David Vala, Vladimir Kasik, Lukas Peter, Silvia Conforto, Maurizio Schmid, Marek Penhaker, Thermal energy harvesting on the bodily surfaces of arms and legs through a wearable thermo-electric generator, *Sensors* 18 (6) (2018) 1927.
- [10] Wei He, Gan Zhang, Xingxing Zhang, Jie Ji, Guiqiang Li, Xudong Zhao, Recent development and application of thermoelectric generator and cooler, *Appl. Energy* 143 (2015) 1–25.
- [11] S.B. Riffat, Xiaoli Ma, Thermoelectrics: a review of present and potential applications, *Appl. Therm. Eng.* 23 (8) (2003) 913–935.
- [12] Dan Dai, Yixin Zhou, Jing Liu, Liquid metal based thermoelectric generation system for waste heat recovery, *Renew. Energy* 36 (12) (2011) 3530–3536.
- [13] Siang Fui Tie, Chee Wei Tan, A review of energy sources and energy management system in electric vehicles, *Renew. Sustain. Energy Rev.* 20 (2013) 82–102.
- [14] K.R. Ullah, R. Saidur, H.W. Ping, R.K. Akikur, N.H. Shuvo, A review of solar thermal refrigeration and cooling methods, *Renew. Sustain. Energy Rev.* 24 (2013) 499–513.
- [15] Ming-Zhi Yang, Chyan-Chyi Wu, Ching-Liang Dai, Wen-Jung Tsai, Energy harvesting thermoelectric generators manufactured using the complementary metal oxide semiconductor process, *Sensors* 13 (2) (2013) 2359–2367.
- [16] Pin-Hsu Kao, Po-Jen Shih, Ching-Liang Dai, Mao-Chen Liu, Fabrication and characterization of CMOS-MEMS thermoelectric micro generators, *Sensors* 10 (2) (2010) 1315–1325.
- [17] Nesrine Jaziri, Ayda Boughamoura, Jens Müller, Brahim Mezghani, Fares Tounsi, Mohammed Ismail, A comprehensive review of thermoelectric generators: Technologies and common applications, *Energy Rep.* 6 (2020) 264–287.
- [18] Yu-Wei Chen, Chyan-Chyi Wu, Cheng-Chih Hsu, Ching-Liang Dai, Fabrication and testing of thermoelectric CMOS-MEMS microgenerators with CNCs film, *Appl. Sci.* 8 (7) (2018) 1047.
- [19] Piotr M. Markowski, Multilayer thick-film thermoelectric microgenerator based on LTCC technology, *Microelectron. Int.* 33 (3) (2016) 155–161.
- [20] Piotr Markowski, Thick-film photoimageable and laser-shaped arms for thermoelectric microgenerators, *Microelectron. Int.* 28 (3) (2011) 43–50.
- [21] Piotr Markowski, Andrzej Dziedzic, Planar and three-dimensional thick-film thermoelectric microgenerators, *Microelectron. Reliabil.* 48 (6) (2008) 890–896.
- [22] Piotr Markowski, Leszek Straszewski, Andrzej Dziedzic, Sandwich-type three-dimensional thick-film thermoelectric microgenerators, in: 2008 31st International Spring Seminar on Electronics Technology, IEEE, 2008.
- [23] Piotr Markowski, Thermoelectric energy harvester fabricated in thick-film/LTCC technology, in: Dr Agata Skwarek (Ed.), *Microelectron. Int.* 31 (3) (2014) 176–185.
- [24] Trung Nguyen Huu, Toan Nguyen Van, Ono Takahito, Flexible thermoelectric power generator with Y-type structure using electrochemical deposition process, *Appl. Energy* 210 (2018) 467–476.
- [25] S. Hasebe, J. Ogawa, M. Shiozaki, T. Toriyama, S. Sugiyama, H. Ueno, K. Itoigawa, Polymer based smart flexible thermopile for power generation, in: 17th IEEE International Conference on Micro Electro Mechanical Systems. Maastricht MEMS 2004 Technical Digest, IEEE.
- [26] Francisco Suarez, Dishit P. Parekh, Collin Ladd, Daryoosh Vashae, Michael D. Dickey, Mehmet C. Öztürk, Flexible thermoelectric generator using bulk legs and liquid metal interconnects for wearable electronics, *Appl. Energy* 202 (2017) 736–745.
- [27] D.T. Crane, J.W. LaGrandeur, Progress report on BSST-led US department of energy automotive waste heat recovery program, *J. Electron. Mater.* 39 (9) (2009) 2142–2148.
- [28] B. Orr, A. Akbarzadeh, P. Lappas, An exhaust heat recovery system utilising thermoelectric generators and heat pipes, *Appl. Therm. Eng.* 126 (2017) 1185–1190.
- [29] Qimin Cao, Weiling Luan, Tongcai Wang, Performance enhancement of heat pipes assisted thermoelectric generator for automobile exhaust heat recovery, *Appl. Therm. Eng.* 130 (2018) 1472–1479.
- [30] Seyed Alireza Mostafavi, Mojtaba Mahmoudi, Modeling and fabricating a prototype of a thermoelectric generator system of heat energy recovery from hot exhaust gases and evaluating the effects of important system parameters, *Appl. Therm. Eng.* 132 (2018) 624–636.
- [31] G.L. Solbrekken, K. Yazawa, A. Bar-Cohen, Thermal management of portable electronic equipment using thermoelectric energy conversion, in: The Ninth International Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (IEEE Cat. No.04CH37543), IEEE.
- [32] Yu Zhou, Somnath Paul, Swarup Bhunia, Harvesting wasted heat in a microprocessor using thermoelectric generators: Modeling, analysis and measurement, in: 2008 Design, Automation and Test in Europe, IEEE, 2008.

- [33] Mingjie Guan, Kunpeng Wang, Dazheng Xu, Wei-Hsin Liao, Design and experimental investigation of a low-voltage thermoelectric energy harvesting system for wireless sensor nodes, *Energy Convers. Manage.* 138 (2017) 30–37.
- [34] Mohamed Al Musleh, Evangelia Topriska, Lynne Jack, David Jenkins, Thermoelectric generator experimental performance testing for wireless sensor network application in smart buildings, in: A. Shanableh, M. Maalej, S. Barakat, M. Omar, S. Al-Toubat, R. Al-Ruzouq, K. Hamad (Eds.), *MATEC Web Conf.* 120 (2017) 08003.
- [35] Achraf Amar, Ammar Kouki, Hung Cao, Power approaches for implantable medical devices, *Sensors* 15 (11) (2015) 28889–28914.
- [36] Tom Torfs, Vladimir Leonov, Chris Van Hoof, Bert Gyselinckx, Body-heat powered autonomous pulse oximeter, in: 2006 5th IEEE Conference on Sensors, IEEE, 2006.
- [37] Moritz Thielen, Lukas Sigrist, Michele Magno, Christofer Hierold, Luca Benini, Human body heat for powering wearable devices: From thermal energy to application, *Energy Convers. Manage.* 131 (2017) 44–54.
- [38] Kai Liu, Yunpeng Liu, Zhiheng Xu, Zhengrong Zhang, Zicheng Yuan, Xiao Guo, Zhangang Jin, Xiaobin Tang, Experimental prototype and simulation optimization of micro-radial milliwatt-power radioisotope thermoelectric generator, *Appl. Therm. Eng.* 125 (2017) 425–431.
- [39] Zicheng Yuan, Xiaobin Tang, Zhiheng Xu, Junqin Li, Wang Chen, Kai Liu, Yunpeng Liu, Zhengrong Zhang, Screen-printed radial structure micro radioisotope thermoelectric generator, *Appl. Energy* 225 (2018) 746–754.
- [40] Abu Raihan Mohammad Siddique, Shohel Mahmud, Bill Van Heyst, A review of the state of the science on wearable thermoelectric power generators (TEGs) and their existing challenges, *Renew. Sustain. Energy Rev.* 73 (2017) 730–744.
- [41] Dipak S. Patil, Rachayya R. Arakerimath, Pramod V. Walke, Thermoelectric materials and heat exchangers for power generation – a review, *Renew. Sustain. Energy Rev.* 95 (2018) 1–22.
- [42] Saim Memon, Khawaja Tahir, Experimental and analytical simulation analyses on the electrical performance of thermoelectric generator modules for direct and concentrated quartz-halogen heat harvesting, *Energies* 11 (12) (2018) 3315.
- [43] Joao Ider, Adhimar Oliveira, Rero Marques Rubinger, A robust system for thermoelectric device characterization, *IEEE Trans Instrum. Measur.* 70 (2021) 1–7.
- [44] Rainer Storn, Kenneth Price, *J. Global Optim.* 11 (4) (1997) 341–359.
- [45] Lian Lian Jiang, Douglas L. Maskell, Jagdish C. Patra, Parameter estimation of solar cells and modules using an improved adaptive differential evolution algorithm, *Appl. Energy* 112 (2013) 185–193.
- [46] D.M. Fébba, E.C. Bortoni, A.F. Oliveira, R.M. Rubinger, The effects of noises on metaheuristic algorithms applied to the PV parameter extraction problem, *Sol. Energy* 201 (2020) 420–436.
- [47] Abdulmunaem H. Elarusi, Hassan Fagehi, Hosung Lee, Alaa Attar, Theoretical approach to predict the performance of thermoelectric generator modules, *J. Electron. Mater.* 46 (2) (2016) 872–881.
- [48] Dario Narducci, Do we really need high thermoelectric figures of merit? A critical appraisal to the power conversion efficiency of thermoelectric materials, *Appl. Phys. Lett.* 99 (10) (2011) 102104.
- [49] Nganyang Paul Bayendang, Mohamed Tariq Ekeramodien Kahn, Vipin Balyan, Ivo Draganov, Sivakumar Pasupathi, *A Comprehensive Thermoelectric Generator (TEG) Modelling*, Zenodo, 2020.
- [50] G. Jeffrey Snyder, Alemayouh H. Snyder, Figure of merit ZT of a thermoelectric device defined from materials properties, *Energy Environ. Sci.* 10 (11) (2017) 2280–2283.
- [51] David E. Goldberg, *Genetic Algorithms*, Pearson Education India, 2006.
- [52] Janez Brest, Sao Greiner, Borko Boskovic, Marjan Mernik, Viljem Zumer, Self-adapting control parameters in differential evolution: A comparative study on numerical benchmark problems, *IEEE Trans. Evolut. Comput.* 10 (6) (2006) 646–657.
- [53] Makbul A.M. Ramli, H.R.E.H. Boucekara, Abdulsalam S. Alghamdi, Optimal sizing of PV/wind/diesel hybrid microgrid system using multi-objective self-adaptive differential evolution algorithm, *Renew. Energy* 121 (2018) 400–411.
- [54] Hamed Beirami, Ali Zargar Shabestari, Mohammad Mahdi Zerafat, Optimal PID plus fuzzy controller design for a PEM fuel cell air feed system using the self-adaptive differential evolution algorithm, *Int. J. Hydrogen Energy* 40 (30) (2015) 9422–9434.
- [55] Chi-Hsien Sun, Chien-Ching Chiu, Min-Hui Ho, Ching-Lieh Li, Comparison of dynamic differential evolution and self-adaptive dynamic differential evolution for buried metallic cylinder, *Res. Nondestruct. Eval.* 24 (1) (2013) 35–50.
- [56] Saswata Nandi, M. Janga Reddy, Comparative performance evaluation of self-adaptive differential evolution with GA, SCE and DE algorithms for the automatic calibration of a computationally intensive distributed hydrological model, *H2Open J.* 3 (1) (2020) 306–327.
- [57] Swagatam Das, Sankha Subhra Mullick, P.N. Suganthan, Recent advances in differential evolution – an updated survey, *Swarm Evol. Comput.* 27 (2016) 1–30.
- [58] Siti Nooraya Mohd Tawil, Mohd Zulkarnain Zainal, Energy harvesting using TEG and PV cell for low power application, author(s), 2018, <http://dx.doi.org/10.1063/1.5022935>.
- [59] Hee Seok Kim, Weishu Liu, Gang Chen, Ching-Wu Chu, Zhifeng Ren, Relationship between thermoelectric figure of merit and energy conversion efficiency, *Proc. Natl. Acad. Sci.* 112 (27) (2015) 8205–8210.